

## Cover Letter

Dear Dr. Siebes,

I am writing to apply for the Postdoctoral Researcher position on FAIR Research Infrastructures for the Social Sciences and Humanities at the Department of Computer Science, Vrije Universiteit Amsterdam. The role brings together exactly the threads I want to pull on next, building FAIR infrastructure that makes heterogeneous research data findable and interoperable, combining knowledge representation with applied AI, and doing that work close to the community producing the data. After four years building open-source data infrastructure during my PhD, this is the natural next step.

My PhD at TU Eindhoven was, in practical terms, an infrastructure project. The Brains4Buildings consortium pulled in data from many sources, BMS streams from several vendors, IoT sensors deployed across living labs, weather APIs, occupant self-reports collected through a dedicated iOS app, and historical sub-metering archives, and someone had to make all of it interoperable before any of it could be used for research. I designed and built ComfortOS, an open-source multi-tenant platform that ingests these streams behind a unified connector gateway and a server-driven schema, so new sites and new data sources can be added without changing code. The point is that other researchers in the consortium, often without engineering backgrounds, could find the data they needed, access it in a consistent form, and rebuild experiments months later. Translating that pattern into the SSH context, with controlled vocabularies, persistent identifiers, and entity linking layered on top, is exactly the work I want to do at VU.

My background is well suited to the technical core of the role. I work in Python every day, have shipped FastAPI, PostgreSQL, and Docker services in production both during my PhD and now at Spectral in Amsterdam, and have published peer-reviewed research alongside the software in venues like *Journal of Building Engineering*, *Energy Efficiency*, *Automation in Construction*, and *Computer Graphics Forum*. My M.Sc. in Mathematics is heavy in graph theory and combinatorial optimisation, which I co-developed into production algorithms at the GPLAN research startup, including variable-chromosome genetic algorithms for Steiner-tree routing on CNN-classified satellite imagery. Graphs are how I naturally think about structured information, so the move into RDF, SKOS, and SPARQL is one I am keen to make, and I have already begun studying the linked data stack. On the GenAI side I have hands-on experience integrating LLMs into research and engineering workflows, and I am especially interested in how retrieval grounded in knowledge graphs and entity linking can hold generative models accountable to curated SSH vocabularies rather than letting them hallucinate over open corpora.

I want to be honest that my primary research output so far has been in the energy and built environment domain rather than SSH. What I bring is the engineering and methodological backbone that FAIR infrastructure work needs, software craftsmanship, multi-source data integration, ML and interpretability, and a habit of building things that other researchers actually use. The most rewarding parts of my PhD were the conversations with non-engineering domain experts, listening for the question behind the question, and building tools that genuinely answered it. I would bring that posture to working with historians, archivists, sociologists, and linguists in the consortium.

Concretely, I would contribute on three fronts. First, engineering the data layer, schemas, vocabularies, and entity-linking services that let SSH datasets become first-class linked-data citizens. Second, building lightweight, reproducible tooling around the data so partners and external researchers can run their own analyses without reinventing the plumbing, drawing on the ComfortOS pattern. Third, exploring GenAI applications that sit responsibly on top of those knowledge graphs, using retrieval over curated vocabularies and provenance-aware prompting to produce outputs that domain experts can audit.

The teaching and consortium component suits me as well. During my PhD I supervised Master's students through full data projects, ran workshops at CAADRIA and BITS Pilani on optimisation, and tutored Smart Building Methodology and Technology at TU/e. Working inside a national consortium with TU Delft, TNO, and industry partners taught me how to keep multi-stakeholder projects moving without losing scientific rigour, which maps closely to the dynamics across the three postdoc positions at VU.

I would welcome the chance to discuss how my experience with open-source research infrastructure, graph methods, ML, and applied software engineering can contribute to the team and to the wider FAIR Linked Data agenda at VU. Thank you for your consideration.

Kind regards,  
Nitant Upasani